

Smart Metro Ticket Booking System

Project Planning Phase

2. Project Logic

1. Purpose

This document captures the core business and system logic that powers the metro digital ticketing solution — specifically the fare calculation logic and the QR ticket generation and validation logic. These logics sit at the heart of the Fare Logic Script and QR Code Integration milestones from the project plan.

2. Fare Calculation Logic

Fare calculation is triggered once a passenger submits Source, Destination and Passenger Type on the Service Catalog form. The logic below shows the step-by-step decision flow used to arrive at the final fare, including concession handling and multi-ticket bookings.

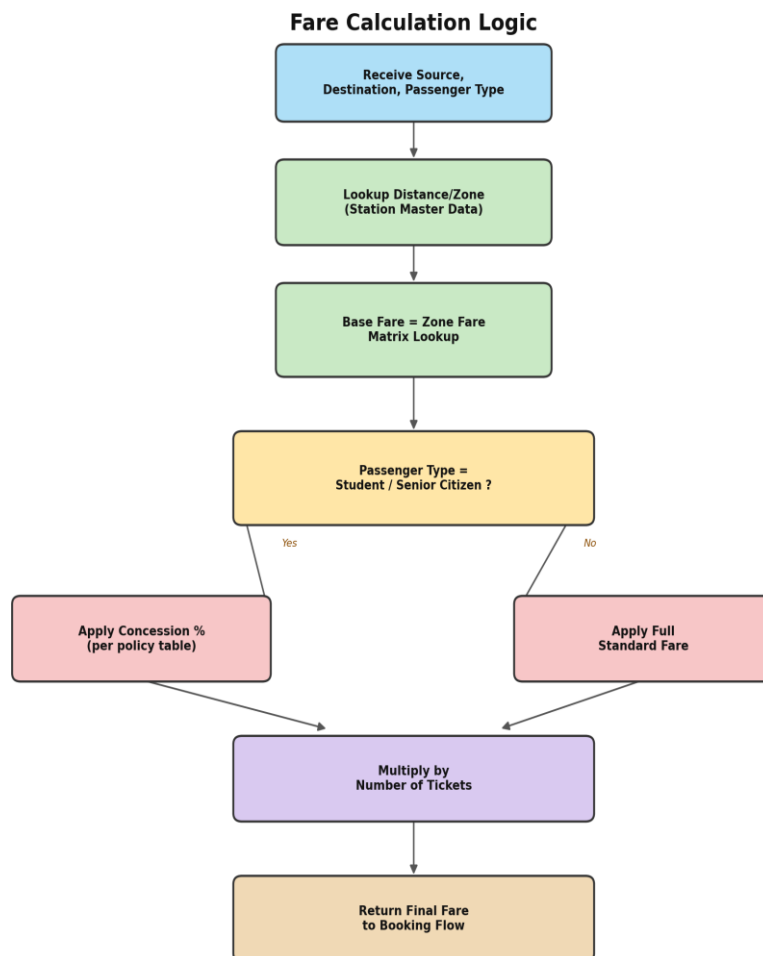


Figure 1 — Fare Calculation Logic Flow

2.1 Fare Logic Rules

Rule	Condition	Action
Base Fare	Always	Looked up from Zone Fare Matrix using Source/Destination distance.
Student Concession	Passenger Type = Student	Apply approved student discount % to base fare.
Senior Citizen Concession	Passenger Type = Senior Citizen	Apply approved senior citizen discount % to base fare.
General Fare	Passenger Type = General	Apply full standard fare with no discount.
Bulk Booking	Number of Tickets > 1	Multiply the resulting per-ticket fare by ticket count.
Invalid Route	Source = Destination	Reject request and prompt passenger to re-select stations.

3. QR Ticket Generation & Validation Logic

Once payment is confirmed, the system moves through a defined lifecycle for the QR ticket — from generation, to active/unused status, to gate scans at entry and exit, and finally to closure and reconciliation. Any duplicate or expired scan attempt is rejected.

QR Ticket Generation & Validation Logic

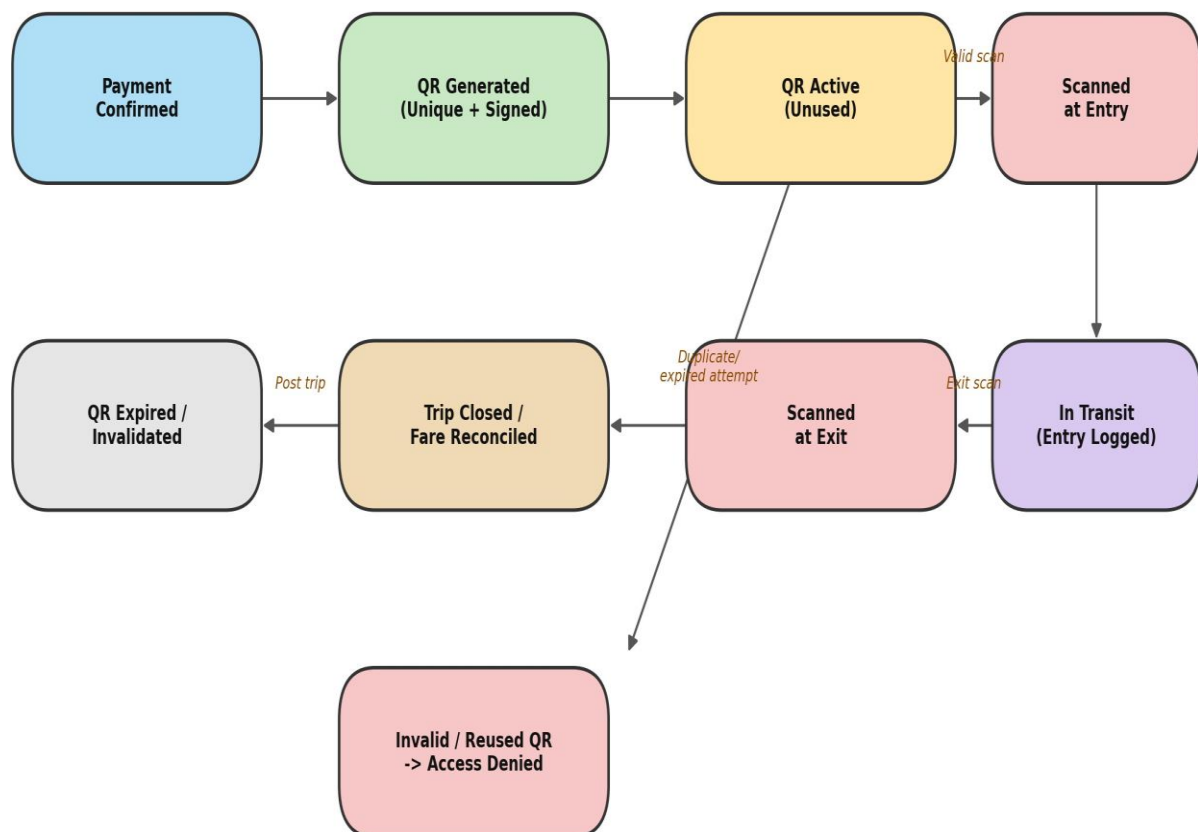


Figure 2 — QR Ticket Generation & Validation State Logic

3.1 QR Lifecycle States

State	Trigger	Next State
Payment Confirmed	Successful payment gateway response	QR Generated
QR Generated	Unique, signed QR created	QR Active (Unused)
QR Active (Unused)	Passenger has ticket, not yet travelled	Scanned at Entry / Expired
Scanned at Entry	Valid QR scanned at AFC entry gate	In Transit
In Transit	Passenger inside metro network	Scanned at Exit
Scanned at Exit	Valid QR scanned at AFC exit gate	Trip Closed / Fare Reconciled
Invalid / Reused QR	Duplicate or expired scan attempt	Access Denied

4. Exception Handling Logic

- **Payment failure** — booking held in a pending state; passenger prompted to retry or cancel; no QR is generated until payment succeeds.
- **Duplicate QR scan** — gate rejects the scan, logs the attempt, and alerts the Station Manager for manual review.
- **Expired QR** — QR codes not used within the validity window are automatically invalidated and cannot be scanned.
- **Concession mismatch** — if passenger type cannot be validated (e.g., missing ID), system falls back to General fare and flags the transaction for review.
- **Integration timeout** — if the QR generation API does not respond within the defined threshold, the booking flow retries automatically before escalating to IT Admins.

5. Summary

Together, the fare calculation logic and QR lifecycle logic ensure that every booking results in an accurate fare, a secure and traceable digital ticket, and a smooth entry-to-exit passenger experience, while giving Operations Teams and IT Admins the controls needed to handle exceptions safely.